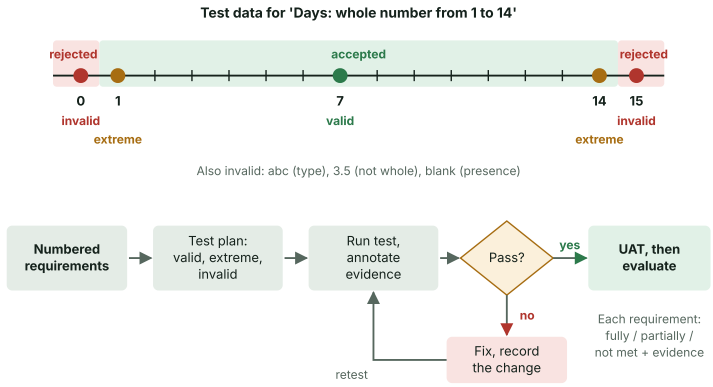
Testing and evaluation against the requirements

AD05 FACT FILE

Specification: students should be able to

- Produce a test plan mapped to the requirements using valid, invalid and extreme (boundary) data
- Carry out testing with annotated evidence of results, corrective action and retesting
- Carry out user acceptance testing
- Evaluate the solution against each requirement as fully, partially or not met, with evidence and justified improvements

Key words: test plan, valid data, invalid data, extreme data, boundary, expected result, actual result, corrective action, retest, user acceptance testing, evaluation, fully met



Choosing valid, extreme and invalid test data for a 1 to 14 range, and the cycle from requirements through testing, fixing and retesting to user acceptance testing and evaluation

Test plans and test data

A test plan lists each test with a number, the requirement it checks, its purpose, the test data, the type of data and the expected result. Columns for the actual result, evidence reference and action taken are completed during testing. Valid data is typical data that should be accepted. Extreme or boundary data lies at the edges of the acceptable range and should also be accepted. Invalid data, including values just outside the range, wrong types and blanks, should be rejected with a helpful message.

Evidence and user acceptance testing

Each test must be evidenced, usually with annotated screenshots showing the data entered and the actual result, cross-referenced by test number. When a test fails, the cause, the correction and a successful retest should all be shown. User acceptance testing involves the client or an end user performing realistic tasks with the finished system, then giving feedback on whether it meets their needs. Their feedback, ideally recorded on a form, provides independent evidence for the evaluation.

Evaluation

The evaluation judges the finished system against every numbered requirement, stating whether each is fully, partially or not met. Each judgement must be supported by evidence, such as test numbers, figures and user feedback, rather than opinion. Partially met and unmet requirements should be explained honestly. Improvements should be realistic, specific and justified, explaining how they would be implemented and how they would benefit the client. Limitations of the solution, such as scalability or remote access, should also be discussed.

T Key terms from the mark schemes

Test plan
A planned list of tests, each with a number, requirement, purpose, test data, data type and expected result.

Extreme data
Data at the boundaries of the acceptable range, which the system should accept; also called boundary data.

User acceptance testing
Testing by the client or end users with realistic tasks to confirm the system meets their requirements.

Valid data
Normal data within the acceptable range that the system should accept.

Invalid data
Data that is outside the acceptable range or of the wrong type, which the system should reject.

Evaluation
A judgement of how well the finished system meets each requirement, supported by evidence, with justified improvements.

Exam focus

A common weakness is testing only validation; test calculations, queries, reports, navigation and non-functional requirements too. Always test both boundaries and the values just outside them. Expected results must be precise: 'error message' is weaker than quoting the message. In the evaluation, avoid descriptive statements such as 'I made a booking form'. Instead write evaluative ones: 'R6 is fully met because tests 1 to 5 show values 1 to 14 are accepted and 0, 15 and text are rejected'.

? Check yourself

1. A field must hold a whole number of days from 1 to 14. Give one example each of valid, extreme and invalid test data. [3]
2. State four columns that should appear in a test plan. [4]
3. Explain why a test plan should be mapped to the numbered requirements. [2]

Answers: 1. Valid (normal): any whole number from 2 to 13, e.g. 5 or 10 (1) / Extreme (boundary): 1 or 14 (1) / Invalid: e.g. 0, 15, -3, 2.5 or 'abc' (1). 2. Any four from: test number (1) / requirement number (1) / purpose / description of test (1) / test data (1) / type of test data (valid, extreme, invalid) (1) / expected result (1) / actual result (1) / evidence reference (e.g. figure number) (1) / action taken / pass or fail (1). 3. So that every requirement is shown to be tested by at least one test and none is missed (1) / and the test results can be used as evidence in the evaluation to judge each requirement as fully, partially or not met (1).